Harrison He

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EDUCATION

University of Illinois Urbana-Champaign Expected May 2030
B.S. in Mathematics (transferring into Mathematics & Computer Science)

St John's College, Johannesburg Graduated 2025
Cambridge A-Levels: A Mathematics, A* Further Mathematics, A Physics, A Chinese, B Chemistry*

PROJECTS & EXPERIENCE

Founder & Developer — 店小力 Dianxiaoli, AI sales agent deployed for paying merchants Apr 2026 – Present · Remote
Architecture, agent design and evaluation write-up: harrisonhe.com/dianxiaoli.html

- Designed, built and operate, as a solo developer, an autonomous agent inside WeChat Customer Service and WhatsApp: it quotes from the merchant's own catalogue, negotiates, records orders, handles payment screenshots and escalates edge cases to the owner. **Live 24/7 for 3 paying merchants** in China and South Africa since April 2026.
- **Safety enforced in code, not prompts:** a wrong price or a falsely confirmed payment ends a merchant relationship, so the hard constraints (no invented prices, no stock counts, no cross-customer prices, never confirm a payment) live in the pipeline: a post-model price guard voids any reply containing a price absent from the retrieved catalogue rows; payments advance only by owner action on an append-only proof ledger; a nightly audit replays production transcripts through 9 checks.
- **Single source of truth for a store (Sep 2026):** multi-tenant routing by channel account; a staff role with per-field visibility (staff see the authorised floor, never cost or other customers' prices), price-change approval and enquiry tickets; four industry packs (wholesale stall / retail / factory / service) switched with one owner command; consent-gated merchant-health export (17 fields).
- **Engineering and evaluation:** deterministic-first pipeline — a rule engine resolves SKU and quantity and the LLM handles only the unmatched path; one model call per turn emits tagged actions (`[ORDER]`, `[WAITLIST]`, `[ESCALATE]`) validated by a deterministic executor; owner notifications on background workers off the request path (P95 < 3 s); **18 automated suites / 1,055 checks** gate every release (blocked my own releases twice). Replaced the first deployment's vector retrieval with exact identifier matching (similarity cannot distinguish adjacent model numbers). Python / FastAPI on Railway; per-merchant deployment cut from 3 weeks to 2–3 days.

External Collaboration — Du Xiaoman Financial, AI Talent Programme (小满摘星计划) Sep 2026 – Present · Remote

- Selected for Du Xiaoman's (Baidu's fintech arm) AI talent programme to develop a standardised, multi-tenant edition of Dianxiaoli for its small-business merchants. Lead product and engineering with two programme-assigned contributors; delivered the merchant-health data interface for the partner's risk models, the multi-tenant base and the industry packs ahead of the agreed roadmap.

Incoming Intern — Allan Gray, Johannesburg Dec 2026

- Confirmed internship at one of South Africa's largest independent asset managers (role and dates to be confirmed).

AI Study Tutor — handwritten past-paper grading 2025 – Present

- Built an LLM pipeline (vision OCR → grading model → step-by-step explanation and new-problem generation) that marks a student's handwritten answer by logical validity, accepting any correct method. Python / FastAPI backend on Railway, Next.js frontend on Vercel.
- **Audited model output by hand against a full paper: 9 of 15 AI marks were false negatives, all in the same direction.** Turned that failure mode into a 13-case regression suite that gates every change.

Cambridge Research Project — regression analysis of cryptocurrency price series St John's College

Actuarial Job Shadows — Discovery; Argen Actuarial Solutions Apr – May 2024 · Johannesburg

- Observed pricing models, risk assessment and actuarial valuations.

ACHIEVEMENTS

- **South African Mathematics Olympiad:** 4th (2021), 7th (2022), Top 50 (2023), Top 20 (2024, 2025) nationally; South Africa IMO selection training camp, Stellenbosch University, 2021–2024.
- **University of the Witwatersrand Mathematics Competition:** Top 10, 2023–2025. South Africa Team Maths Competition: 2nd place 2021–2024, 4th 2025. Estonian Maths Open: Top 20–50 within South Africa.
- South African Informatics Olympiad: first round passed 2023, 2024. Qualified Professional Level 1 Go player (China). 100+ hours volunteer tutoring in high-school mathematics and olympiad preparation, 2022–2025.

SKILLS

Programming: Python (FastAPI, pandas), JavaScript / HTML / CSS (Next.js), Java, Git, Playwright; deployment and operations on Railway and Vercel.

AI & data engineering: LLM agent design with DeepSeek, Anthropic and OpenAI APIs — tagged-action output, deterministic executors, rule-engine fallback, background workers, multi-tenant routing, role-based field visibility; evaluation gates, regression suites and transcript audits; consent-gated data export; WeChat and WhatsApp platform integration.

Languages: Mandarin and English (both fluent).